MIL-R-27208/11B(USAF)
5 October 1973
SUPERSEDING
MIL-R-27208/11A(USAF)
13 December 1968
(See 6.3)

MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND

(ADJUSTMENT TYPE, LEAD-SCREW ACTUATED),

STYLE RT27

* THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.3).

1. SCOPE

1.1 This specification covers the detail requirements for style RT27, adjustment type, leadscrew actuated, wire-wound, variable resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect, on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-27208 - Resistors, Variable, Wire-Wound (Adjustment Type) General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

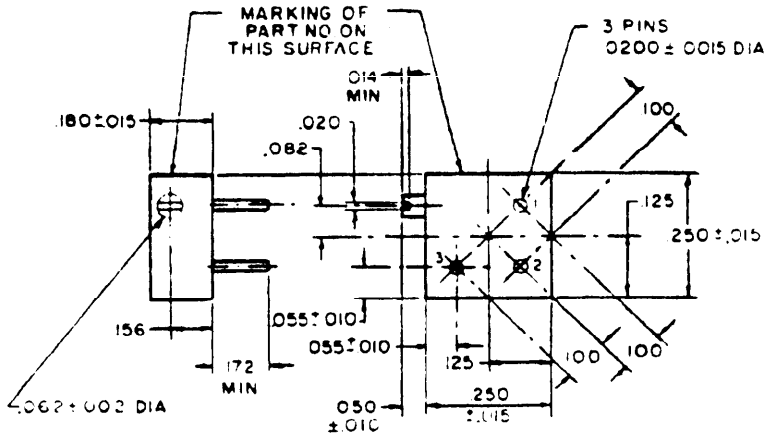
3.1 Requirements. Requirements shall be in accordance with MIL-R-27208, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure.

3.3 Power rating. The power rating shall be 3/8 watt, based on full-load operation at an ambient temperature of 85°C.

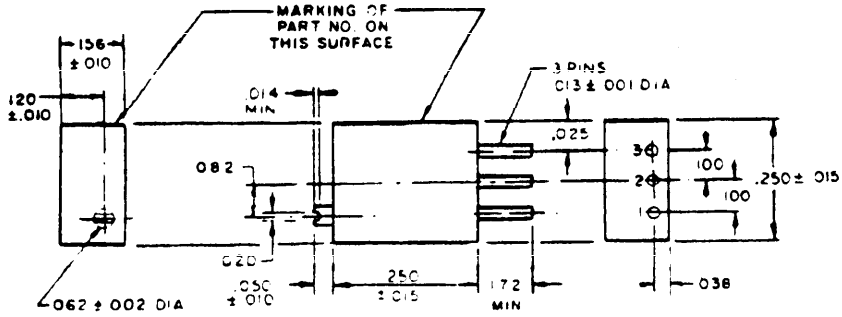
3.4 Nominal resistance value, maximum resolution and maximum rated ac or dc working voltage. Nominal resistance value, maximum resolution, and maximum rated ac or dc working voltage are as specified in table.

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TERMINAL TYPE P

INCHES	MM
.0015	.04
.002	.05
.010	.25
.014	.36
.015	.38
.020	.51
.050	1.27
.055	1.40
.062	1.57
.082	2.08
.100	2.54
.125	3.18
.156	3.96
.172	4.37
.180	4.57
.250	6.35

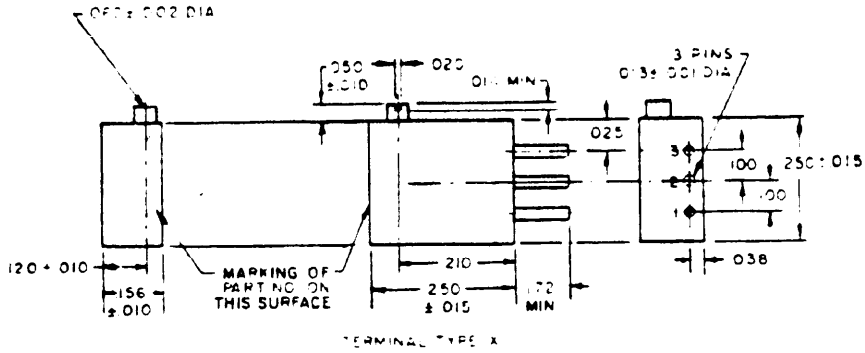


TERMINAL TYPE W

INCHES	MM
.001	.03
.002	.05
.010	.25
.013	.33
.014	.36
.015	.38
.020	.51
.025	.64
.038	.97
.050	1.27
.062	1.57
.082	2.08
.100	2.54
.120	3.05
.156	3.96
.172	4.37
.250	6.35

Style RT27 resistors.

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TERMINAL TYPE X

INCHES	MM
.001	.03
.002	.05
.010	.25
.013	.33
.014	.36
.015	.38
.020	.51
.025	.64
.038	.97
.050	1.27
.062	1.57
.100	2.54
.120	3.05
.156	3.96
.172	4.37
.210	5.33
.250	6.35

NOTES
1. Dimensions are in inches.
2. Metric equivalents to the nearest .01 mm are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .005 (.13 mm).
4. The entire slot of the actuating screw must be above the surface of the unit.

Style RT07 resistors - Continued.

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Nominal resistance value, maximum resolution and maximum rated ac or dc working voltage.

Nominal resistance value	Maximum resolution 3/	Maximum rated ac or dc working voltage
(ohms)	(percent)	(volts)
10	1.85	1.93
20	1.50	2.70
50	1.23	4.30
100	0.90	6.12
200	0.68	8.60
500	0.65	13.70
1,000	0.57	19.36
2,000 1/	0.44	27.05
5,000 2/	0.35	43.30

1/ Value based on the use of wire having no less than a 0.001-inch nominal diameter.

2/ Value based on the use of wire having no less than 0.0008-inch nominal diameter.

3/ Maximum resolutions shown are theoretical.

3.5 Actual effective electrical travel. Actual effective electrical travel shall be 15 turns minimum and 30 turns maximum.

3.6 Operating torque. Operating torque shall be a maximum of 5.0 ounce-inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-27208.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential shall be 600 volts rms at atospheric pressure, and 250 volts rms at reduced barometric pressure.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-27208.

6. NOTES

6.1 Notes. The notes specified in MIL-R-27208 are applicable to this specification.

* 6.2 Supersession data. This specification supersedes MIL-R-27208/11A(USAF), 13 December, 1968.

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* 6.3 Substitution data. The resistors specified herin are not for use in design after date of this specification. They are authorized for use in design contracts effective prior to or on the date of this specification, and to support existing military equipment. There is no ER (MIL-R-39015) replacement for this style.

6.4 Weight. The maximum weight is approximately .6 gram.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 17, 80

Preparing Activity:
Air Force - 85

(Project 5905-F245)

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OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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